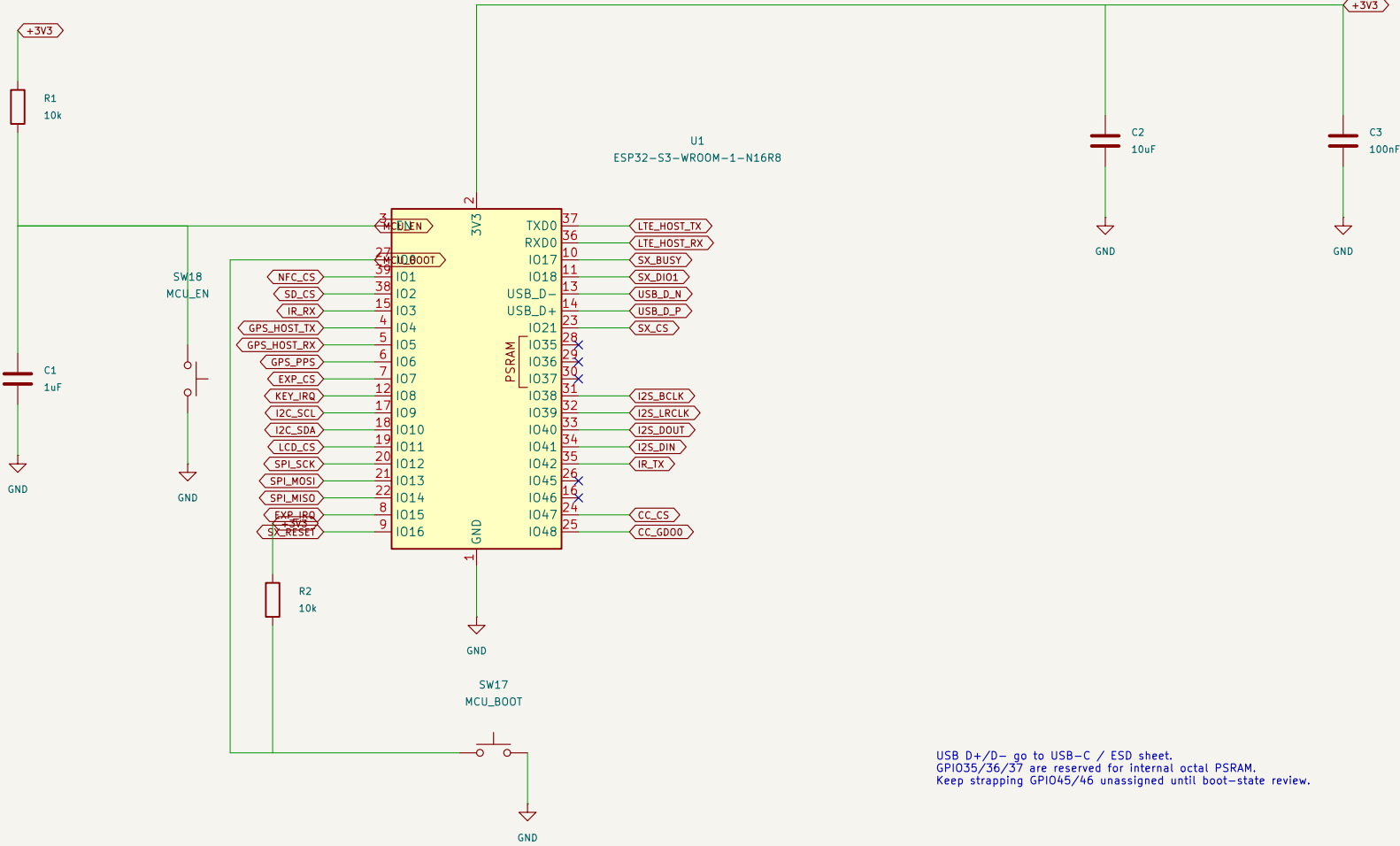


HANDSET REV A / SYSTEM SCHEMATIC INDEX																																											
Engineering draft. Retained ESP32–S3 / MakerFocus 3000 mAh / external display / all peripheral interfaces.																																											
A	ESP32 host, boot and reset			Key matrix and trackball			USB–C data and ESD																																				
	File: core.kicad_sch			File: keys.kicad_sch			File: usb.kicad_sch																																				
B	Fused USB input and eFuse			USB–C source detection and permission			Battery, thermistor and programmable charger																																				
	File: usb–input.kicad_sch			File: usb–detect.kicad_sch			File: power.kicad_sch																																				
C	Power control and fuel gauge			Main 3.3 V buck–boost supply			Auxiliary 5 V and 1.8 V supplies																																				
	File: power–control.kicad_sch			File: reg3.kicad_sch			File: aux–power.kicad_sch																																				
D	Regulated modem supply and enable			SX1262 LoRa radio			CC1101 sub–GHz radio																																				
	File: modem–power.kicad_sch			File: radios.kicad_sch			File: cc–radio.kicad_sch																																				
E	125 kHz reader analog circuit			LF reader logic translation			GNSS receiver and passive antenna																																				
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F	NFC reader, oscillator and local supplies			Media speaker amplifier and digital microphone			Cellular adapter and call–audio harnesses																																				
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G	Protected external module port			MicroSD and case–mounted infrared			External display, touch and backlight																																				
	File: protection.kicad_sch			File: storage–ir.kicad_sch			File: display.kicad_sch																																				
H	Always–on charging supervisor																																										
	File: power–supervisor.kicad_sch																																										
Sheet: / File: handset.kicad_sch <table><tr><td colspan="10">Title:</td><td></td></tr><tr><td colspan="5">Size: A2</td><td colspan="5">Date:</td><td>Rev:</td></tr><tr><td colspan="5">KiCad E.D.A. 10.0.6</td><td colspan="5"></td><td>Id: 1/23</td></tr></table>											Title:											Size: A2					Date:					Rev:	KiCad E.D.A. 10.0.6										Id: 1/23
Title:																																											
Size: A2					Date:					Rev:																																	
KiCad E.D.A. 10.0.6										Id: 1/23																																	

ESP32 host, boot and reset

Retain N16R8. Module pads 28/29/30 belong to octal PSRAM. USB programming uses the USB sheet.



USB D+/D- go to USB-C / ESD sheet.
GPIO35/36/37 are reserved for internal octal PSRAM.
Keep strapping GPIO45/46 unassigned until boot-state review.

RESET / EN: 10k pull-up + 1uF
BOOT held low during reset selects download mode.

Circuit review in progress – not released for fabrication

Sheet: /ESP32 host, boot and reset/
File: core.kicad_sch

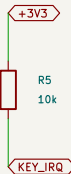
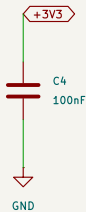
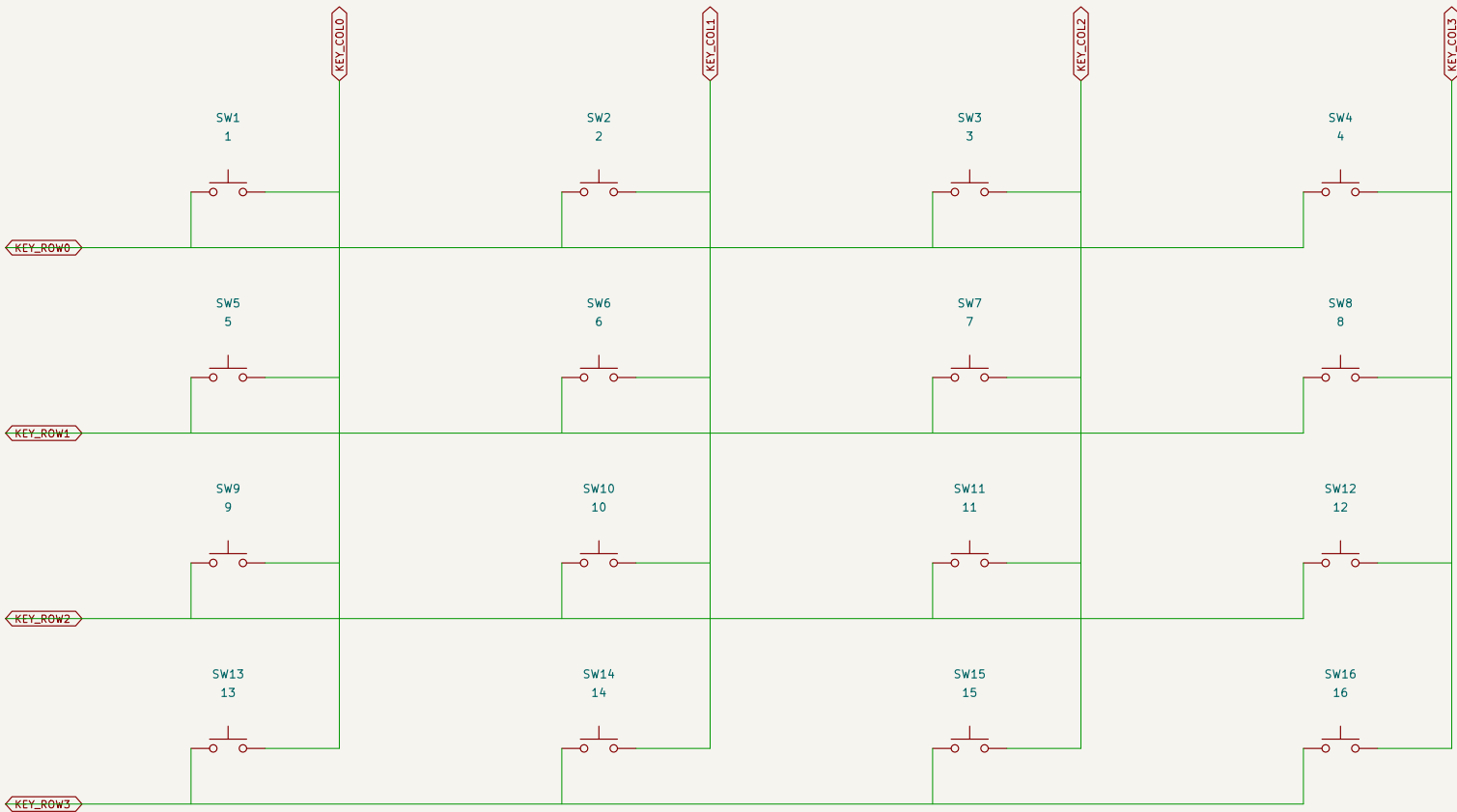
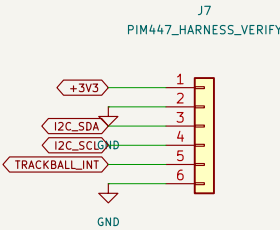
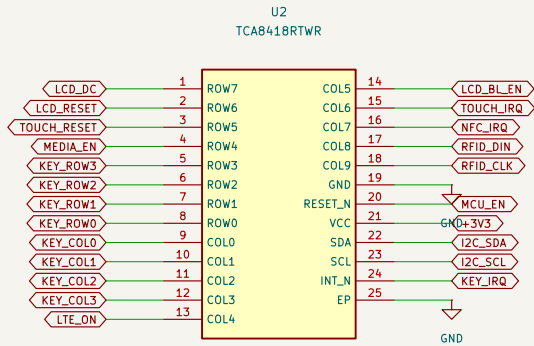
Title: ESP32 host, boot and reset

Size: A3
Date: 2026-09-24

Rev: A – engineering draft
Id: 2/23

Key matrix and trackball

TCAB418: 4 rows x 4 columns; remaining pins are firmware-configured GPIO. Confirm PIM447 harness orientation.



Circuit review in progress – not released for fabrication

Sheet: /Key matrix and trackball/
File: keys.kicad_sch

Title: Key matrix and trackball

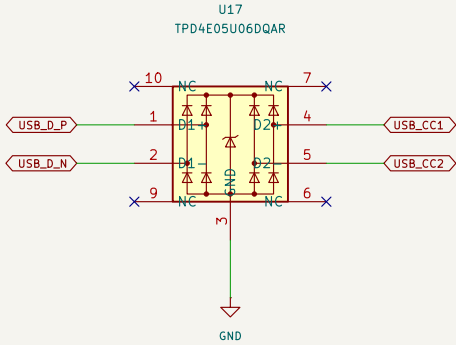
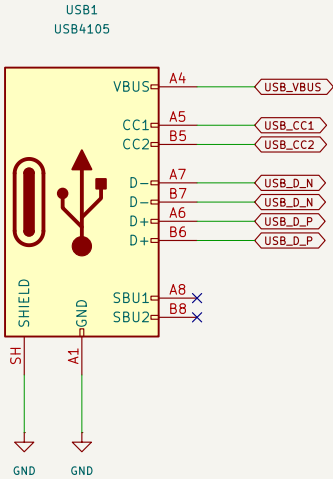
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 3/23

USB-C data and ESD

CC1/CC2 terminate at the sink controller on usb-detect. Data pair goes to the ESP32. Shield/ground return needs layout review.



Circuit review in progress – not released for fabrication

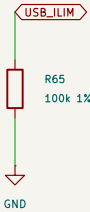
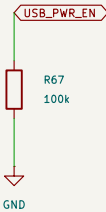
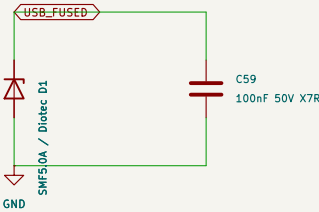
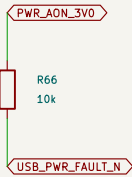
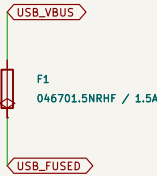
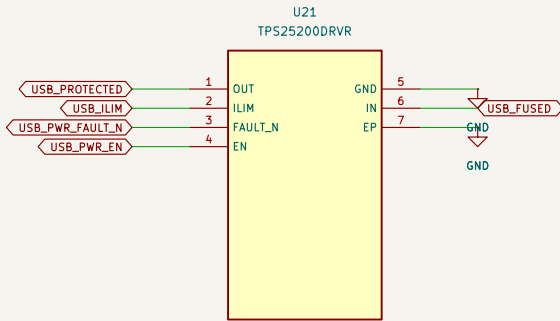
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File: usb.kicad_sch

Title: USB-C data and ESD

Size: A3	Date: 2026-09-24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 4/23

Fused USB input and eFuse

TPS25200 R_ILIM = 100k, 1%. F1/TVS pulse coordination requires bench validation. No USB PD negotiation.



Supply flag: external USB VBUS through F1. No internal source is implied.

Circuit review in progress – not released for fabrication

Sheet: /Fused USB input and eFuse/
File: usb-input.kicad_sch

Title: Fused USB input and eFuse

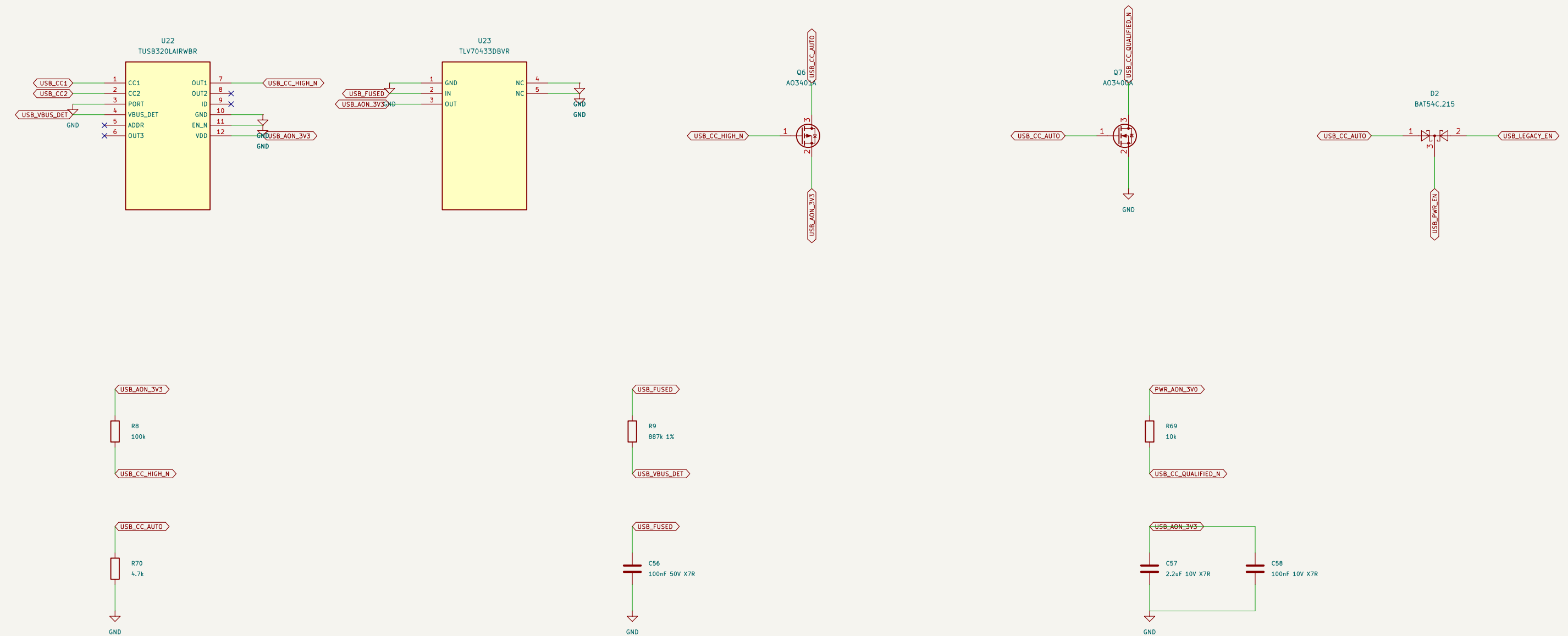
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 5/23

USB-C source detection and permission

Sink-only internal Rd. OUT1 low authorizes Type-C 1.5A/3A. Legacy permission requires USB enumeration; see charger-policy.json.



Circuit review in progress – not released for fabrication

Sheet: /USB-C source detection and permission/
File: usb-detect.kicad_sch

Title: USB-C source detection and permission

Size: A3

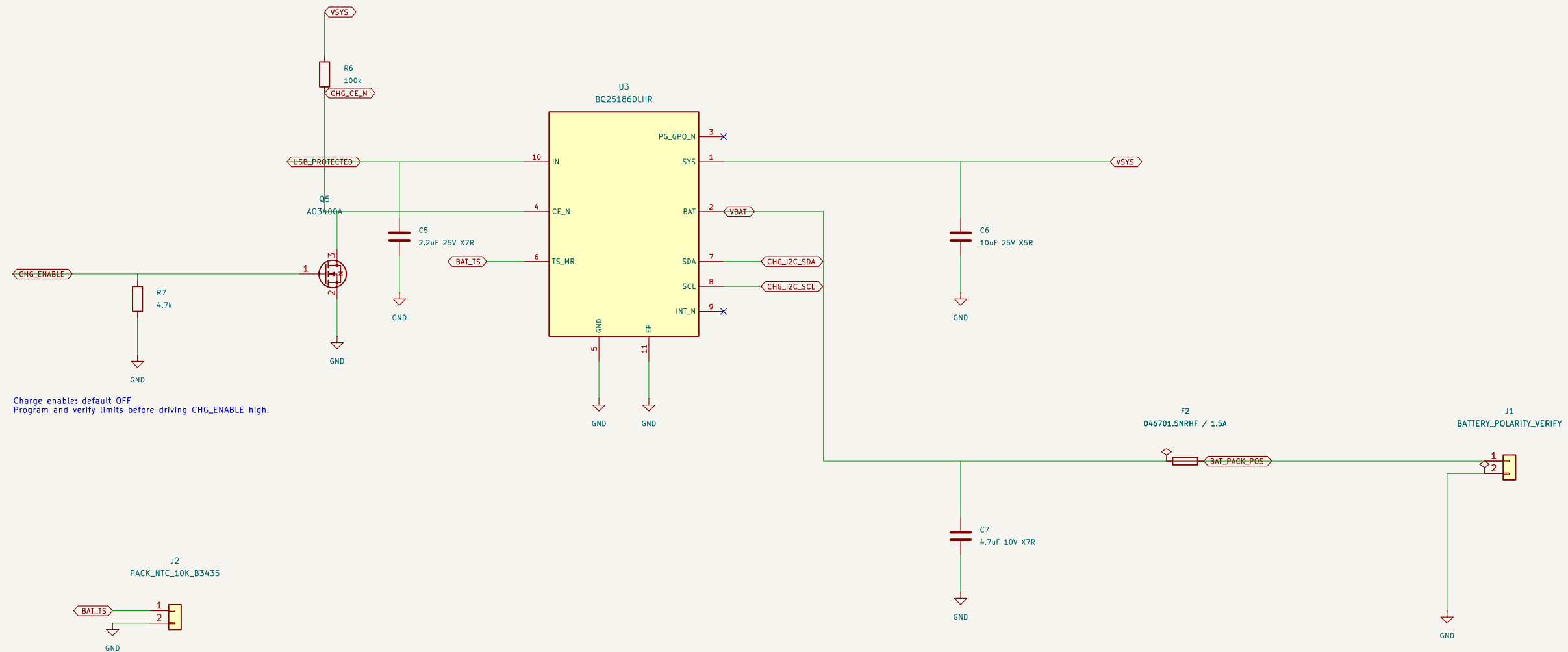
Date: 2026-09-24

Rev: A – engineering draft

Id: 6/23

Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.



Protected 103665 pack / 3000 mAh
F2 does not enforce the pack operating-current limit.

Supply flags identify the external protected pack through F2 and its return.

Circuit review in progress – not released for fabrication

Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

Title: Battery, thermistor and programmable charger

Size: A3

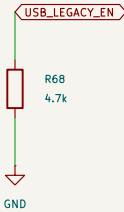
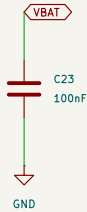
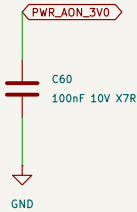
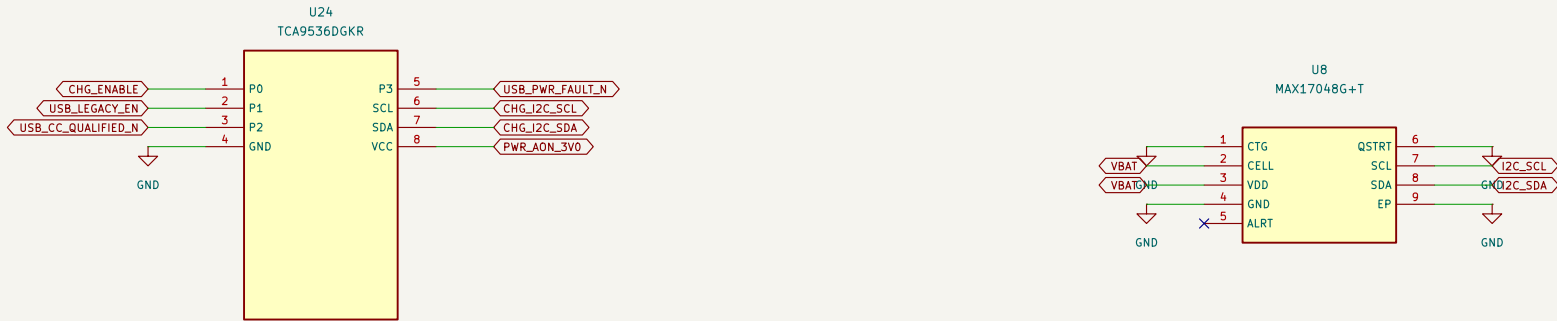
Date: 2026-09-24

Rev: A – engineering draft

Id: 7/23

Power control and fuel gauge

I2C: TCA9536 0x41, MAX17048 0x36, charger 0x6A. Preload low output latches before enabling charger—control outputs.



Circuit review in progress – not released for fabrication

Sheet: /Power control and fuel gauge/
File: power-control.kicad_sch

Title: Power control and fuel gauge

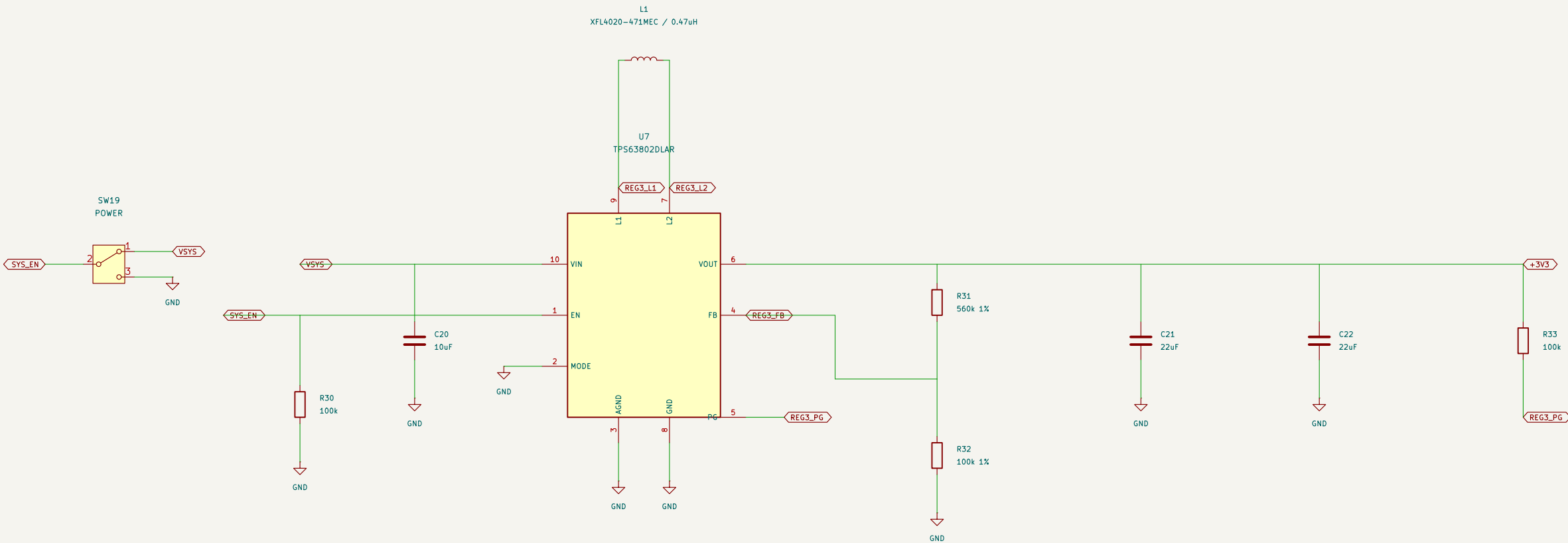
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 8/23

Main 3.3 V buck–boost supply

TPS63802 local PCB copper is routed. External distribution, effective capacitance, thermal limits and load transients remain to qualify.



0.6 V reference x (1 + 560k / 100k) = 3.3 V nominal.
Feedback senses the output capacitor node; keep it clear of the switching loop.

SW19 controls SYS_EN.
Battery / charger circuitry can remain energized while logic rails are off.

Circuit review in progress – not released for fabrication

Sheet: /Main 3.3 V buck–boost supply/
File: reg3.kicad_sch

Title: Main 3.3 V buck–boost supply

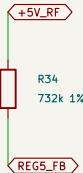
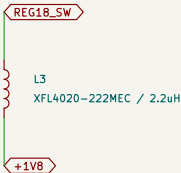
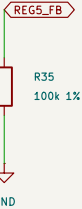
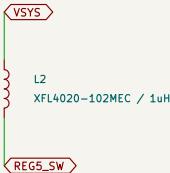
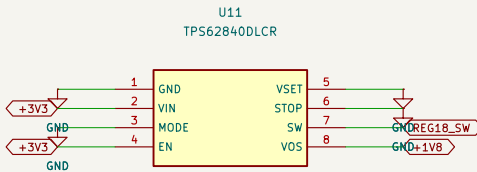
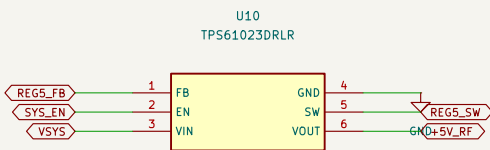
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026–09–24

Rev: A – engineering draft
Id: 9/23

Auxiliary 5 V and 1.8 V supplies

Switching loops must be short in PCB layout. The 5 V rail also supplies display backlight and the LF reader.



Circuit review in progress – not released for fabrication

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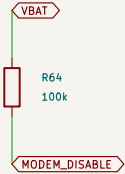
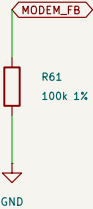
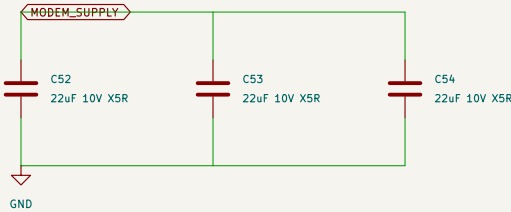
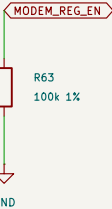
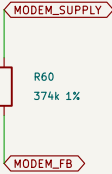
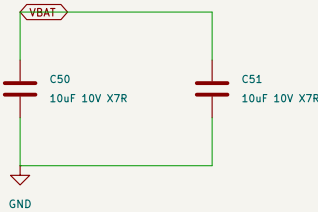
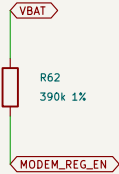
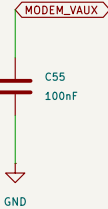
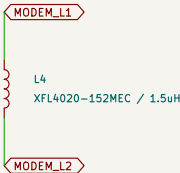
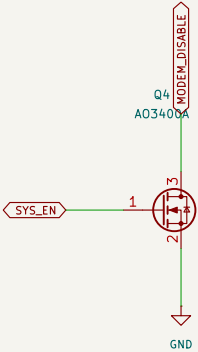
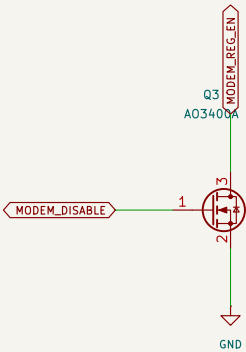
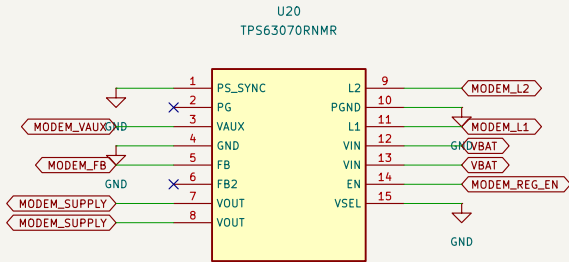
Title: Auxiliary 5 V and 1.8 V supplies

Size: A3
Date: 2026-09-24

Rev: A – engineering draft
Id: 10/23

Regulated modem supply and enable

Nominal output 3.792 V. Exact modem burst current and MakerFocus pack current budget are unresolved release blockers.



Circuit review in progress – not released for fabrication

Sheet: /Regulated modem supply and enable/
File: modem-power.kicad_sch

Title: Regulated modem supply and enable

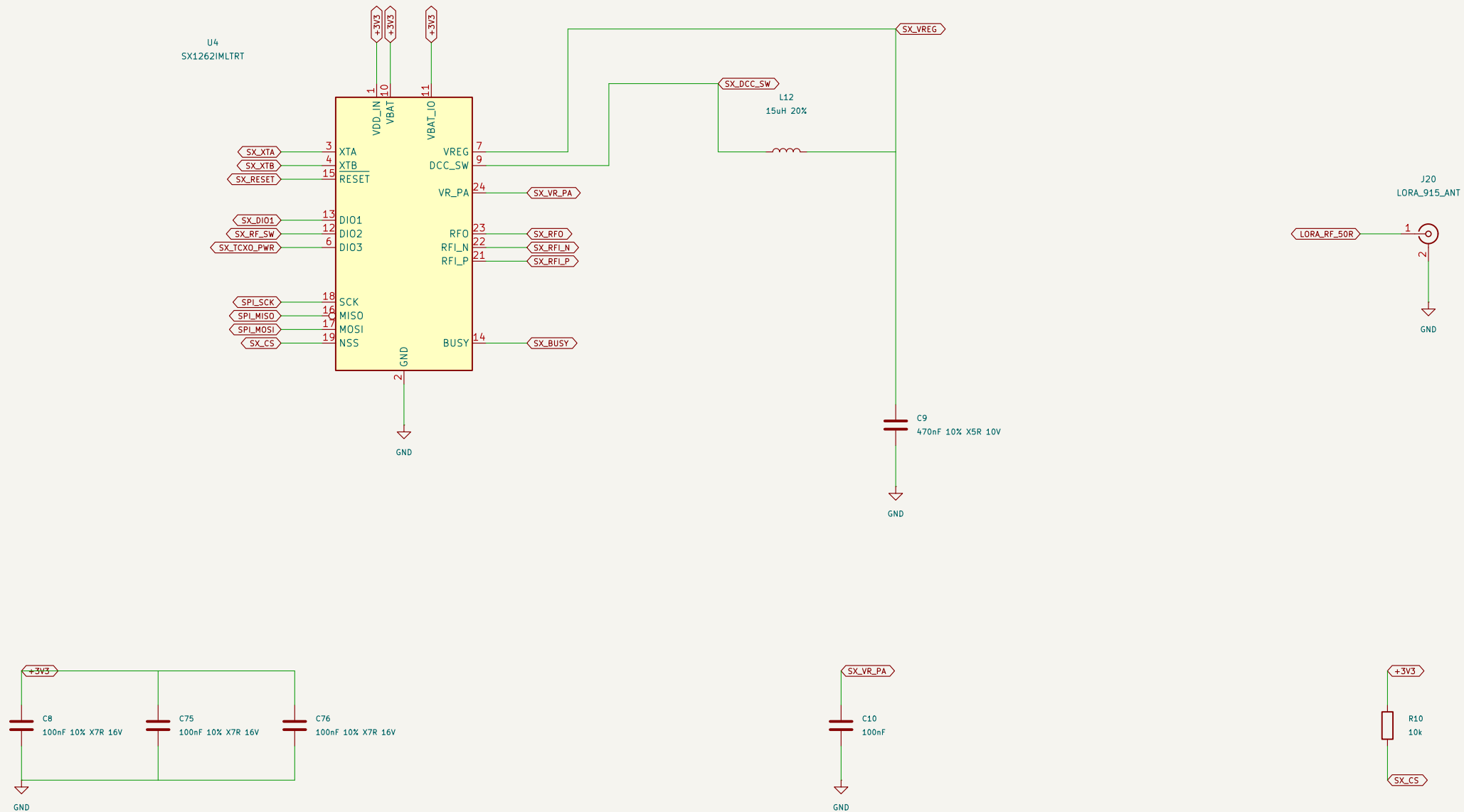
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 11/23

SX1262 LoRa radio

DC-DC support captured from Semtech DS rev1.2 Fig14-2 and Table5-3. Clock and RF frontend remain incomplete. Do not power.



C8: VBAT pin10. C75: VDD_IN pin1. C76: VBAT_IO pin11.
All use the same +3V3 rail; place each close to its pin.
C9 replaces the incorrect 100nF with Semtech reference 470nF.

SetRegulatorMode(0x01) only in STDBY_RC, after hardware qualification.
L12 supplies the core, not the SX1262 high-power PA.
TCXO, PA feed, matching/filter and RF switch are still missing.
C10 PA bypass and J20 antenna connection remain provisional.

Circuit review in progress – not released for fabrication

Sheet: /SX1262 LoRa radio/
File: radios.kicad_sch

Title: SX1262 LoRa radio

Size: A3 Date: 2026-09-24

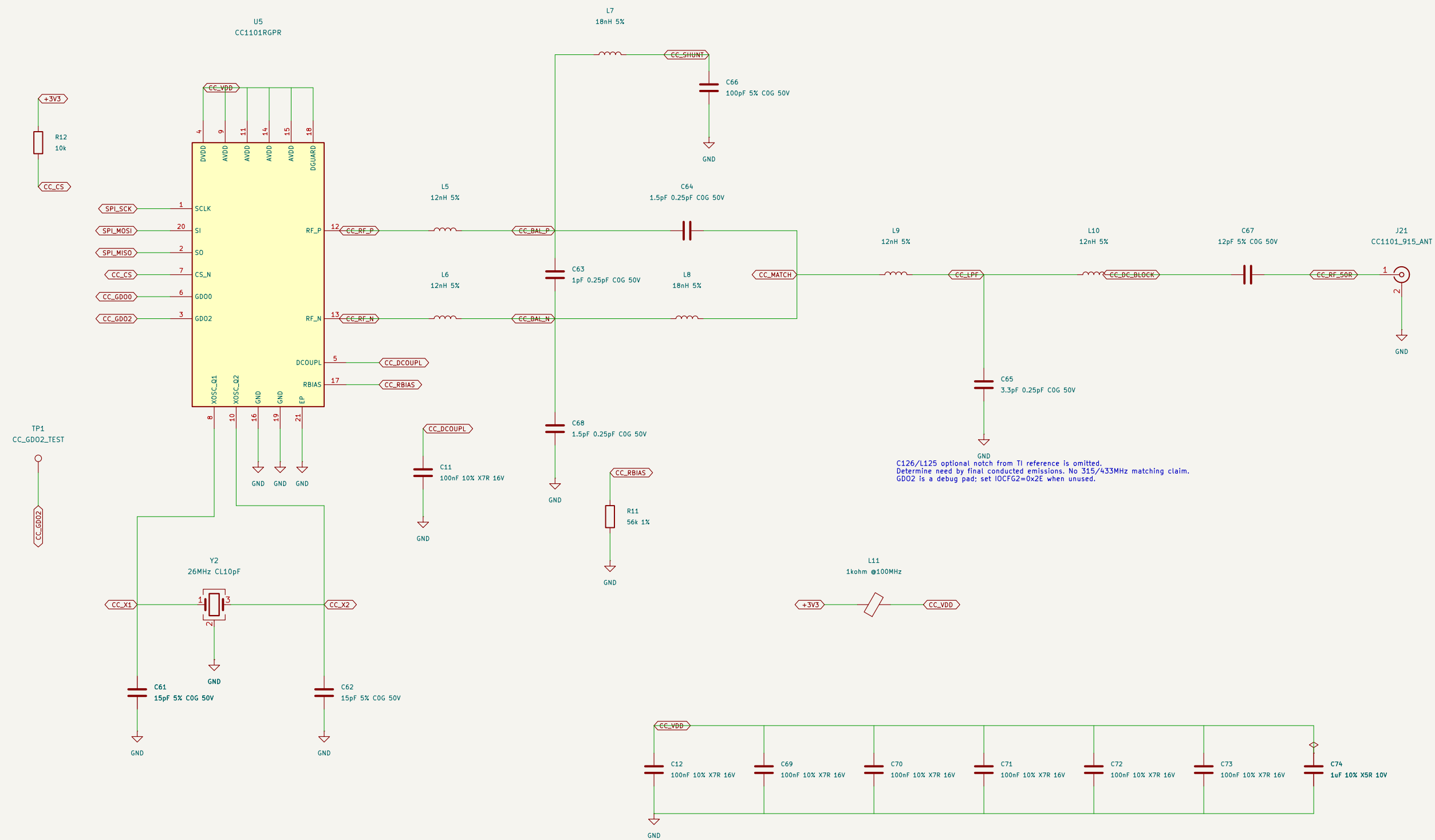
KiCad E.D.A. 10.0.6

Rev: A – engineering draft

Id: 12/23

CC1101 sub-GHz radio

868/915MHz frontend: TI SWRR045 rev3.0 / SWRS061I Fig11. RF layout and tuning are NOT qualified. Separate 50-ohm antenna.



Y2: ABM8-26.000MHZ-10-D-1-G-T
CL = 15pF / 2 + estimated 2.5pF stray = 10pF.
Measure oscillator frequency, startup and drive level.

CC_VDD flag: main +3V3 through passive L11; not an independent source.
Place one 100nF at each supply pin group, plus local 1uF bulk.

Circuit review in progress – not released for fabrication

Sheet: /CC1101 sub-GHz radio/
File: cc-radio.kicad_sch

Title: CC1101 sub-GHz radio

Size: A3

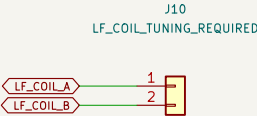
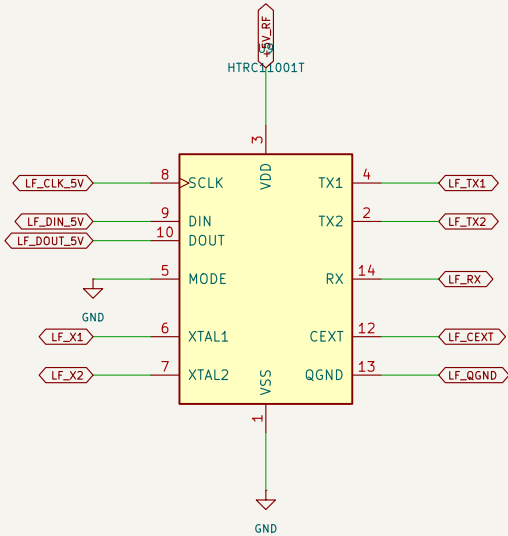
Date: 2026-09-24

Rev: A – engineering draft

Id: 13/23

125 kHz reader analog circuit

HTRC110 requires its oscillator, QGND/CEXT bypass, resonant coil network and a timing-capable host interface.



Circuit review in progress – not released for fabrication

Sheet: /125 kHz reader analog circuit/
File: lf-radio.kicad_sch

Title: 125 kHz reader analog circuit

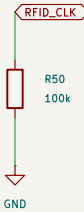
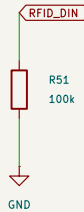
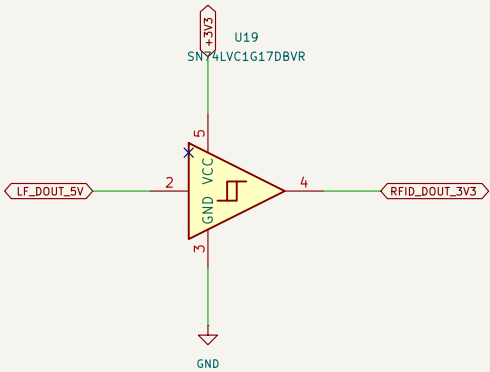
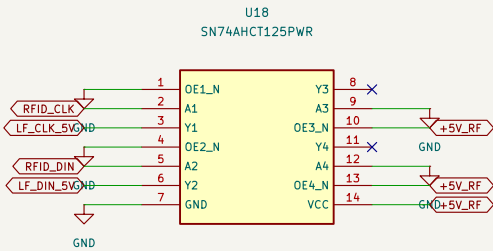
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 14/23

LF reader logic translation

AHCT inputs accept 3.3 V logic at 5 V supply; LVC return buffer is powered at 3.3 V. Host timing remains a design requirement.



Circuit review in progress – not released for fabrication

Sheet: /LF reader logic translation/
File: lf-support.kicad_sch

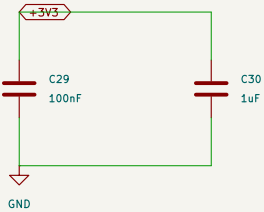
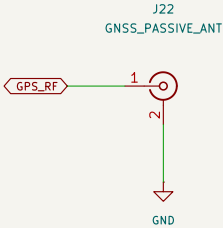
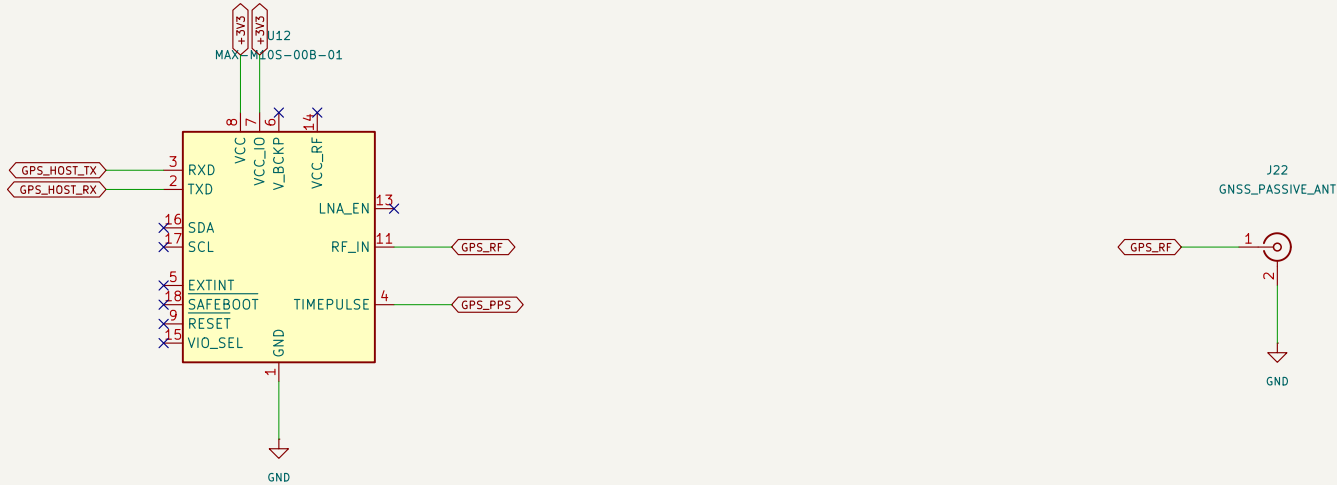
Title: LF reader logic translation

Size: A3
Date: 2026-09-24

Rev: A – engineering draft
Id: 15/23

GNSS receiver and passive antenna

MAX-M10S passive antenna. V_BCKP remains open: cold starts after power removal. No direct LiPo connection.



Circuit review in progress – not released for fabrication

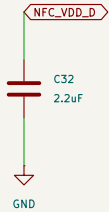
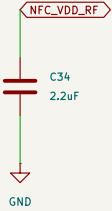
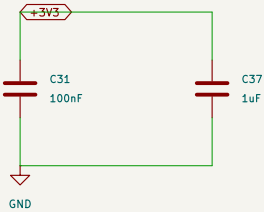
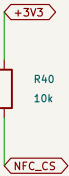
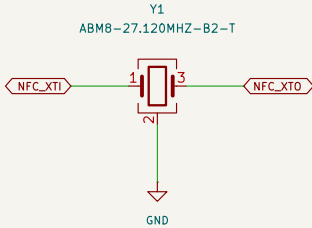
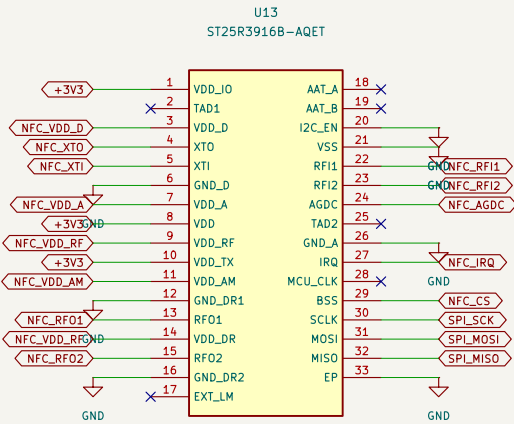
Sheet: /GNSS receiver and passive antenna/
File: gnss-antennas.kicad_sch

Title: GNSS receiver and passive antenna

Size: A3	Date: 2026-09-24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 16/23

NFC reader, oscillator and local supplies

ST25R3916B needs a tuned case-mounted loop and TX/RX matching network. Verify crystal load against the selected crystal.



Circuit review in progress – not released for fabrication

Sheet: /NFC reader, oscillator and local supplies/
File: nfc.kicad_sch

Title: NFC reader, oscillator and local supplies

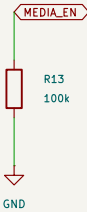
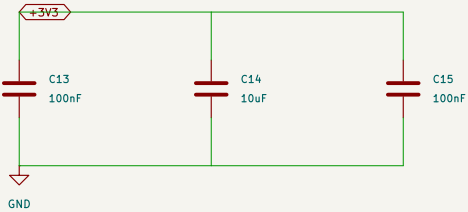
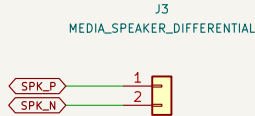
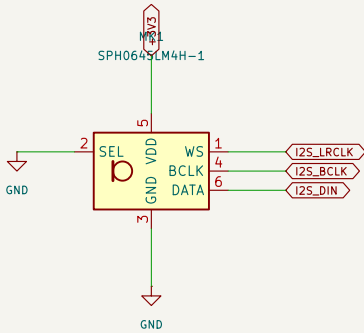
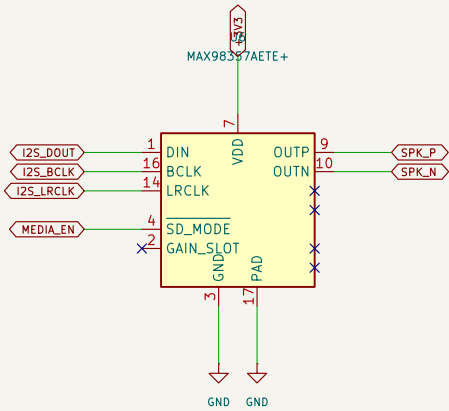
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 17/23

Media speaker amplifier and digital microphone

Top-mounted media speaker uses differential outputs: neither speaker wire is ground. Digital microphone is case-bottom mounted.



Circuit review in progress – not released for fabrication

Sheet: /Media speaker amplifier and digital microphone/
File: audio.kicad_sch

Title: Media speaker amplifier and digital microphone

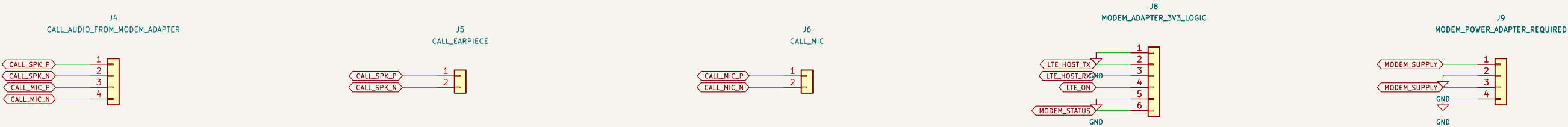
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 18/23

Cellular adapter and call–audio harnesses

Replaceable modem adapter owns SIM, cellular RF and modem–level translation. No universal–carrier guarantee. Speaker top / mic bottom.



Circuit review in progress – not released for fabrication

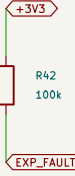
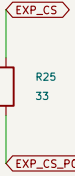
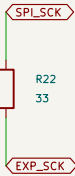
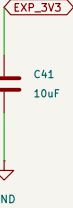
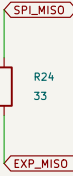
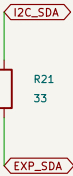
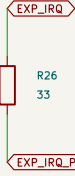
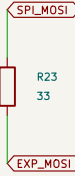
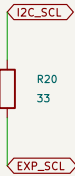
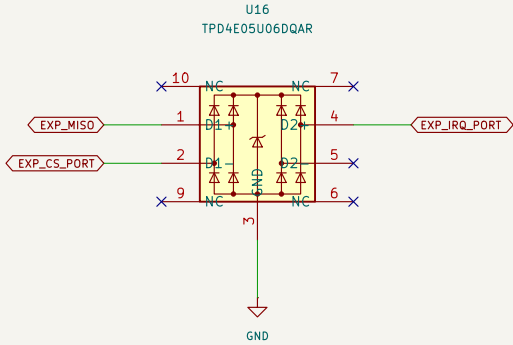
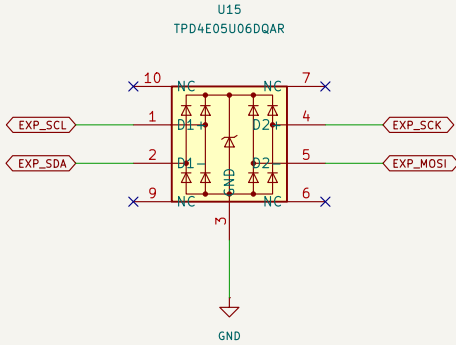
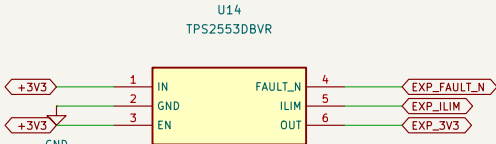
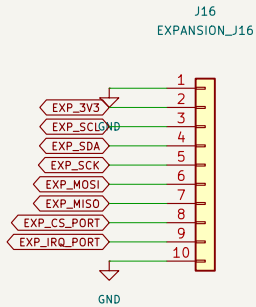
Sheet: /Cellular adapter and call–audio harnesses/
File: interfaces.kicad_sch

Title: Cellular adapter and call–audio harnesses

Size: A3	Date: 2026–09–24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 19/23

Protected external module port

J16 pin order retained. Switch accessories with power off. TPS2553 limit and ESD path require qualification.



Circuit review in progress – not released for fabrication		
Sheet: /Protected external module port/ File: protection.kicad_sch		
Title: Protected external module port		
Size: A3	Date: 2026-09-24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 20/23

MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.

Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress – not released for fabrication		
Sheet: /MicroSD and case-mounted infrared/ File: storage-ir.kicad_sch		
Title: MicroSD and case-mounted infrared		
Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 21/23

MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.

The schematic diagram illustrates the electrical connections for a MicroSD card and a case-mounted infrared (IR) LED and receiver. The components are organized into three main sections: the MicroSD card interface, the IR LED driver, and the IR receiver supply.

MicroSD Card Interface:

- The MicroSD card is connected to the shield via a 10-pin header (J23).
- The card's pins are connected to the shield's pins: SD_DAT2 (1), SD_CS (2), SPL_MOSI (3), +3V3 (4), SPL_SCK (5), VSS (6), DAT0 (7), DAT1 (8), DET_A (9), and DET_B (10).
- The shield's SH pin is connected to GND.

IR LED Driver:

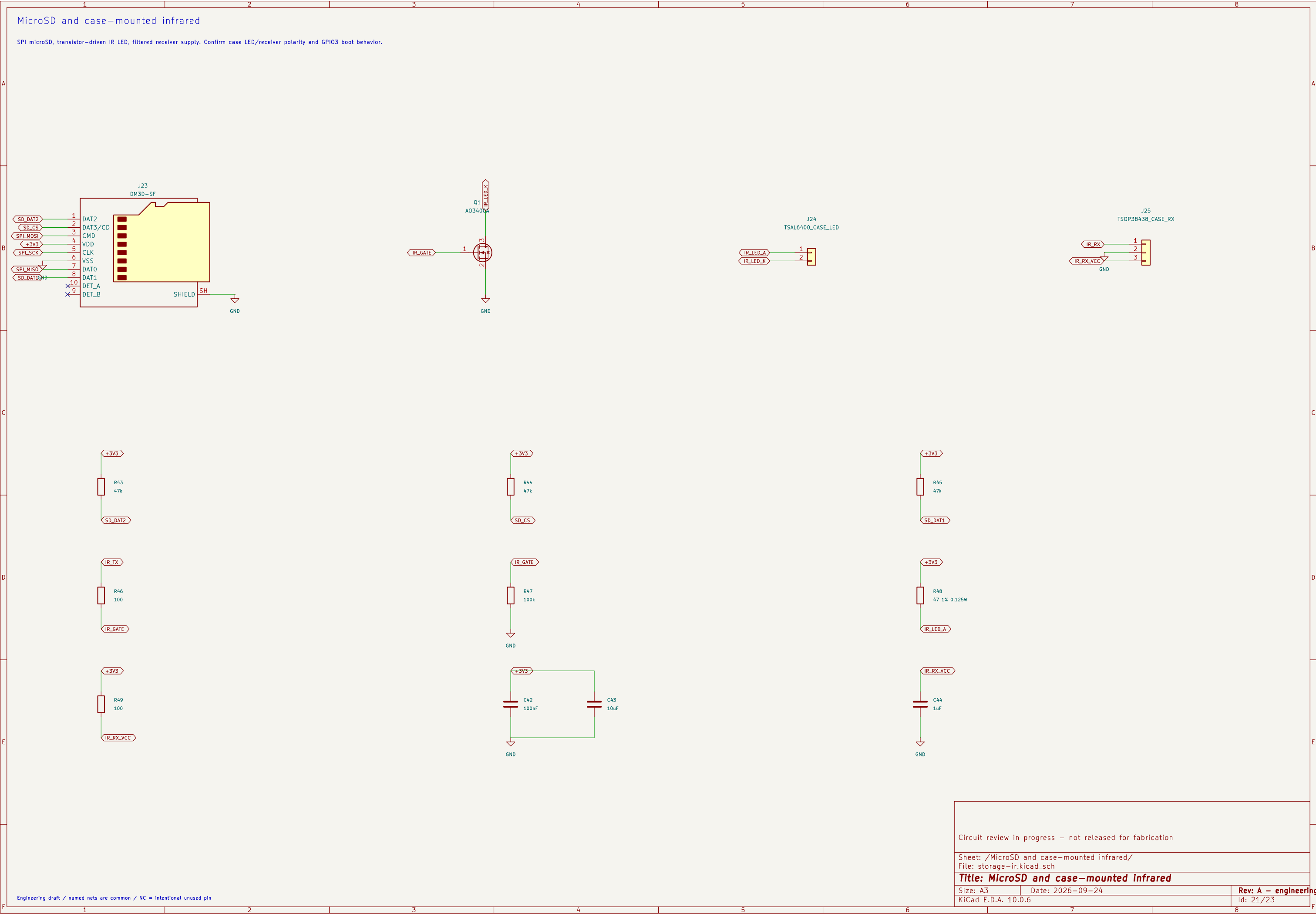
- The IR LED (TSAL6400_CASE_LED) is connected to the shield via a 2-pin header (J24).
- The LED's anode (A) is connected to the shield's IR_LED_A pin, and its cathode (K) is connected to the shield's IR_LED_K pin.
- The IR LED is driven by the shield's IR_GATE pin through a 47k resistor (R43).

IR Receiver Supply:

- The IR receiver (TSOP38438_CASE_RX) is connected to the shield via a 3-pin header (J25).
- The receiver's pins are connected to the shield's pins: IR_RX (1), IR_RX_VCC (2), and GND (3).
- The receiver's supply (IR_RX_VCC) is connected to the shield's IR_RX_VCC pin through a 100nF capacitor (C42).
- The receiver's ground (GND) is connected to the shield's GND pin through a 10uF capacitor (C43).

Power and Grounding:

- The circuit is powered by a 3V3 supply.
- Grounding is established through various pins: SH, IR_LED_K, IR_RX_VCC, and GND.



Sheet: /MicroSD and case-mounted infrared/ File: storage-ir.kicad_sch		
Title: MicroSD and case-mounted infrared		
Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6	Id: 21/23	

MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.

Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress – not released for fabrication		
Sheet: /MicroSD and case-mounted infrared/ File: storage-ir.kicad_sch		
Title: MicroSD and case-mounted infrared		
Size: A3	Date: 2026-09-24	Rev: A – engineering
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MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.

The schematic diagram illustrates the electrical connections for a MicroSD card and a case-mounted infrared (IR) LED and receiver. The components are organized into three main sections: the MicroSD card interface, the IR LED driver, and the IR receiver supply.

MicroSD Card Interface:

- The MicroSD card is connected to the shield via a 10-pin header (J23).
- The card's pins are connected to the shield's pins: SD_DAT2 (1), SD_CS (2), SPL_MOSI (3), +3V3 (4), SPL_SCK (5), VSS (6), DAT0 (7), DAT1 (8), DET_A (9), and DET_B (10).
- The shield's SH pin is connected to GND.

IR LED Driver:

- The IR LED (TSAL6400_CASE_LED) is connected to the shield via a 2-pin header (J24).
- The LED's anode (A) is connected to the shield's IR_LED_A pin, and its cathode (K) is connected to the shield's IR_LED_K pin.
- The IR LED is driven by the shield's IR_GATE pin through a 47k resistor (R43).

IR Receiver Supply:

- The IR receiver (TSOP38438_CASE_RX) is connected to the shield via a 3-pin header (J25).
- The receiver's pins are connected to the shield's pins: IR_RX (1), IR_RX_VCC (2), and GND (3).
- The receiver's supply (IR_RX_VCC) is connected to the shield's IR_RX_VCC pin through a 100nF capacitor (C42).
- The receiver's ground (GND) is connected to the shield's GND pin through a 10uF capacitor (C43).

Power and Grounding:

- The circuit is powered by a 3V3 supply.
- Grounding is established through GND pins and capacitors (C42, C43, C44).

Legend:

- Engineering draft / named nets are common / NC = intentional unused pin

Metadata:

- Sheet: /MicroSD and case-mounted infrared/
- File: storage-ir.kicad_sch
- Title: **MicroSD and case-mounted infrared**
- Size: A3
- Date: 2026-09-24
- Rev: A – engineering
- KiCad E.D.A. 10.0.6
- Id: 21/23

MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.

The schematic diagram illustrates the electrical connections for a MicroSD card, a transistor-driven IR LED, and a filtered receiver supply. The components and their connections are as follows:

- DM3D-SF (J23):** A shield component with pins 1 through 10. Connections include SD_DAT2 (pin 1), SD_CS (pin 2), SPI_MOSI (pin 3), +3V3 (pin 4), SPI_SCK (pin 5), VSS (pin 6), DAT0 (pin 7), DAT1 (pin 8), DET_A (pin 9), and DET_B (pin 10). The SH pin is connected to GND.
- A03400A (Q1):** A transistor used for driving the IR LED. Its base is connected to IR_GATE, its emitter to GND, and its collector to the IR_LED_K pin of the TSAL6400 case LED.
- TSAL6400_CASE_LED (J24):** A two-pin component with pins 1 (IR_LED_A) and 2 (IR_LED_K).
- TSOP38438_CASE_RX (J25):** A three-pin component with pins 1 (IR_RX), 2 (IR_RX_VCC), and 3 (GND).
- Resistors:**
 - R43 (47k): Connected between +3V3 and SD_DAT2.
 - R44 (47k): Connected between +3V3 and SD_CS.
 - R46 (100): Connected between IR_TX and IR_GATE.
 - R47 (100k): Connected between IR_GATE and GND.
 - R48 (47 1% 0.125W): Connected between +3V3 and IR_LED_A.
 - R49 (100): Connected between +3V3 and IR_RX_VCC.
- Capacitors:**
 - C42 (100nF): Connected between +3V3 and GND.
 - C43 (10uF): Connected between +3V3 and GND.
 - C44 (1uF): Connected between IR_RX_VCC and GND.

Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress – not released for fabrication

Sheet: /MicroSD and case-mounted infrared/
File: storage-ir.kicad_sch

Title: MicroSD and case-mounted infrared

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 21/23

MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.

The schematic diagram illustrates the electrical connections for a MicroSD card, a transistor-driven IR LED, and a filtered receiver supply. The components and their connections are as follows:

- DM3D-SF Shield (J23):** A yellow shield component with pins 1 through 10. Pin 1 is connected to SD_DAT2, pin 2 to DAT2, pin 3 to DAT3/CD, pin 4 to CMD, pin 5 to VDD, pin 6 to CLK, pin 7 to VSS, pin 8 to DAT0, pin 9 to DET_A, and pin 10 to DET_B. The SH pin is connected to GND.
- A03400A Transistor (Q1):** A transistor with its base connected to IR_GATE, its emitter to GND, and its collector to the IR_LED_K pin of the TSAL6400 case LED.
- TSAL6400 CASE_LED (J24):** A two-pin component with pins 1 and 2 connected to IR_LED_A and IR_LED_K, respectively.
- TSOP38438 CASE_RX (J25):** A three-pin component with pins 1, 2, and 3 connected to IR_RX, IR_RX_VCC, and GND, respectively.
- Resistors:**
 - R43 (47k) is connected between +3V3 and SD_DAT2.
 - R44 (47k) is connected between +3V3 and SD_CS.
 - R45 (47k) is connected between +3V3 and SD_DAT1.
 - R46 (100) is connected between IR_TX and IR_GATE.
 - R47 (100k) is connected between IR_GATE and GND.
 - R48 (47 1% 0.125W) is connected between +3V3 and IR_LED_A.
 - R49 (100) is connected between +3V3 and IR_RX_VCC.
- Capacitors:**
 - C42 (100nF) and C43 (10uF) are connected in parallel between +3V3 and GND.
 - C44 (1uF) is connected between IR_RX_VCC and GND.

Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress – not released for fabrication

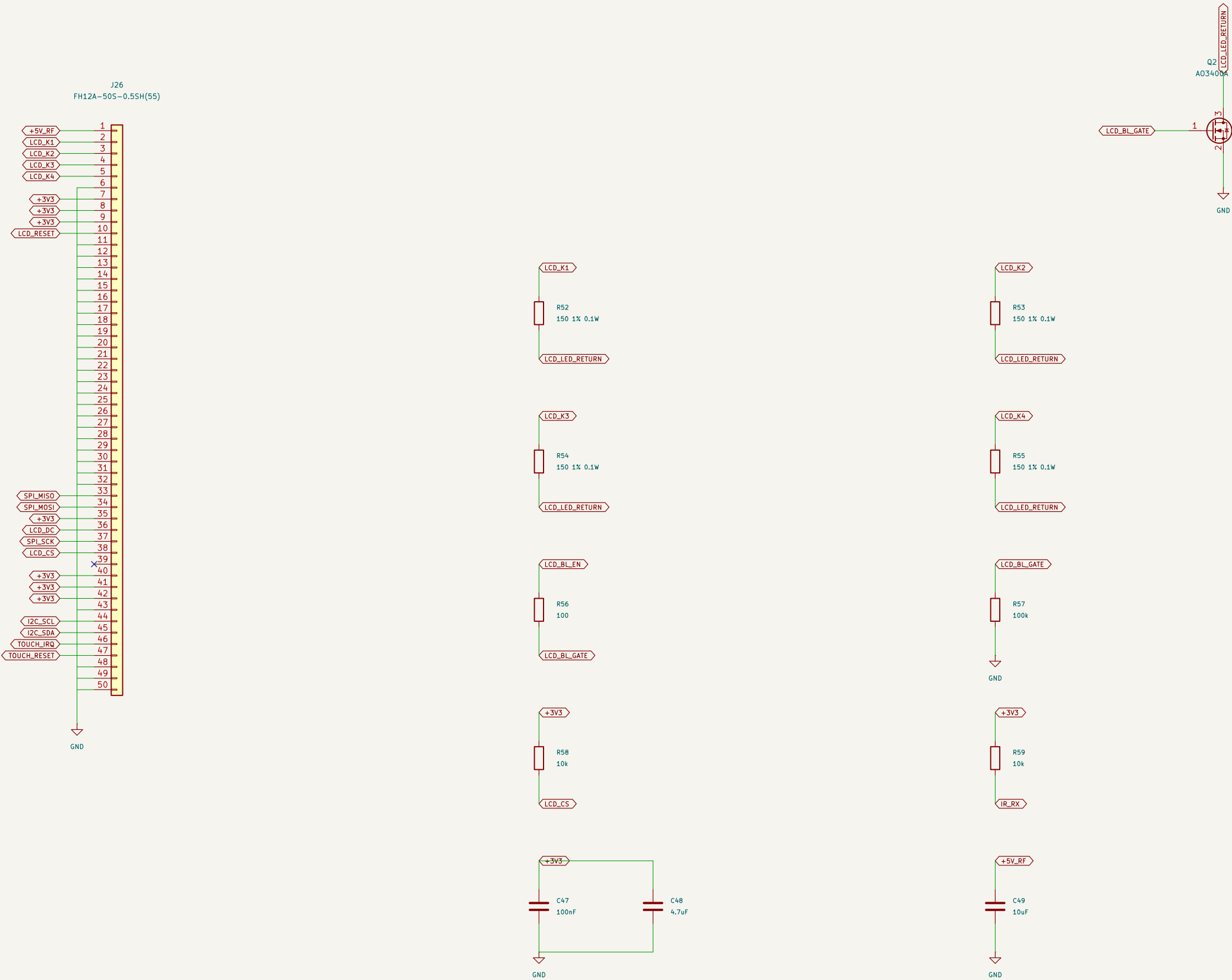
Sheet: /MicroSD and case-mounted infrared/
File: storage-ir.kicad_sch

Title: MicroSD and case-mounted infrared

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 21/23

External display, touch and backlight

ER-TFT024IPS-3 connector contract. Verify purchased CTP option, FPC orientation, LED branches and supply tolerances.



Circuit review in progress – not released for fabrication

Sheet: /External display, touch and backlight/
File: display.kicad_sch

Title: External display, touch and backlight

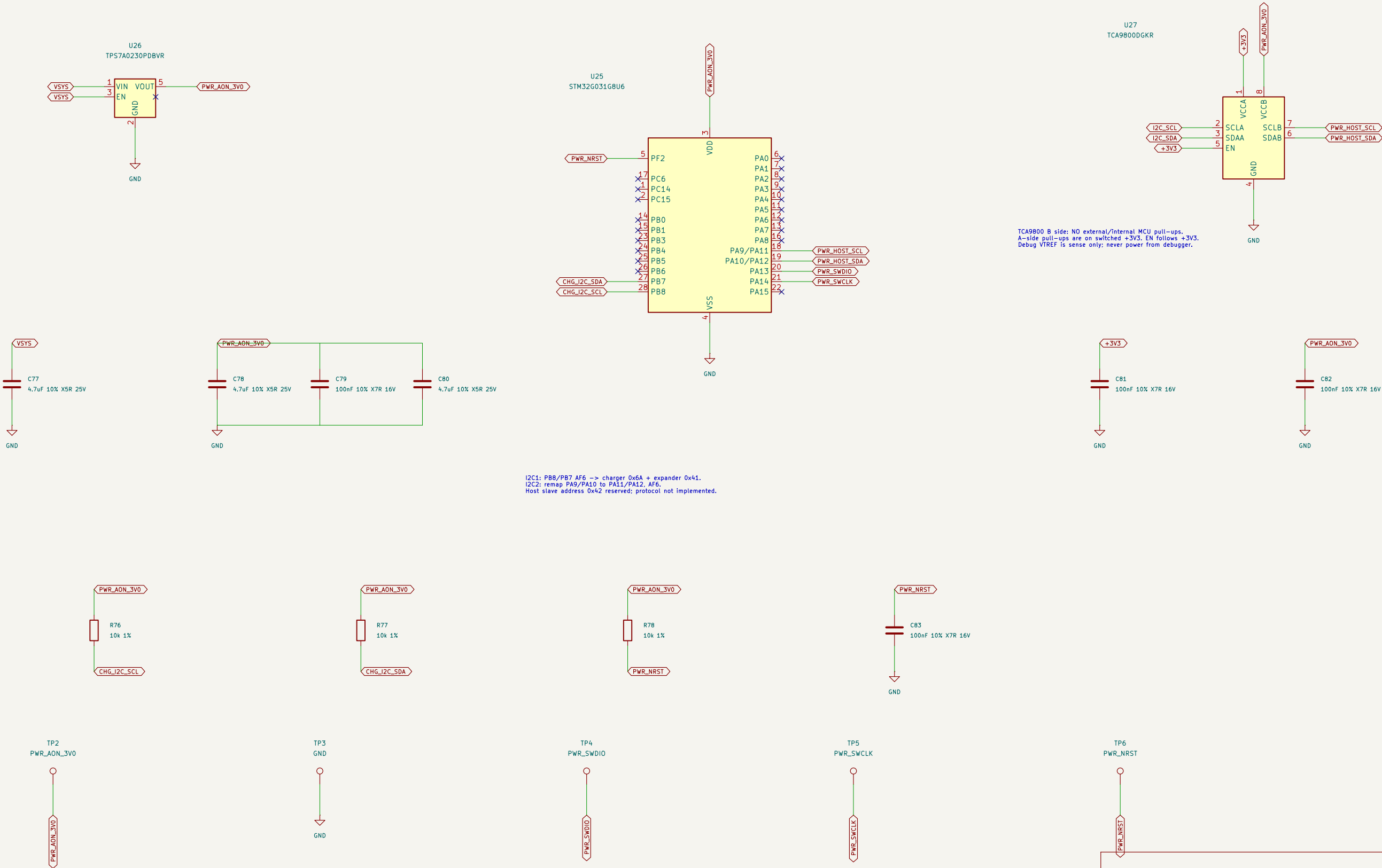
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 22/23

Always-on charging supervisor

Separate supervisor selected. U3 remains the charger. Main host powers off at SW19. Circuit capture only; routing, STM32 firmware and fault qualification pending.



I2C1: PB8/PB7 AF6 -> charger 0x6A + expander 0x41.
I2C2: remap PA9/PA10 to PA11/PA12, AF6.
Host slave address 0x42 reserved; protocol not implemented.

TCA9800 B side: NO external/Internal MCU pull-ups.
A-side pull-ups are on switched +3V3. EN follows +3V3.
Debug VTREF is sense only; never power from debugger.

Circuit review in progress – not released for fabrication

Sheet: /Always-on charging supervisor/
File: power-supervisor.kicad_sch

Title: Always-on charging supervisor

Size: A3 Date: 2026-09-24 Rev: A – engineering draft
KiCad E.D.A. 10.0.6 Id: 23/23